

Limits and Continuity

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Limits

We now study the continuity of functions with two variables. Consider a function $f(x, y)$ and $(a, b) \in \mathbb{R}^2$. We say that the limit of f at (a, b) is some number L if the values of $f(x, y)$ gets arbitrarily close to L as (x, y) approaches (a, b) .

In mathematical terms, we can define the limit as follows:

Definition We say that the limit of $f(x, y)$ as (x, y) approaches (a, b) is L if for every number $\epsilon > 0$ there is a corresponding number $\delta > 0$ such that

$$0 < \sqrt{(x - a)^2 + (y - b)^2} < \delta \implies |f(x, y) - L| < \epsilon$$

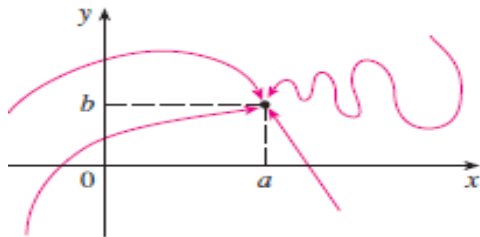
Notation

$$\lim_{(x, y) \rightarrow (a, b)} f(x, y) = L.$$

Recall that for $y = f(x)$, $\lim_{x \rightarrow a} f(x)$ exists if and only if $\lim_{x \rightarrow a^+} f(x)$ and $\lim_{x \rightarrow a^-} f(x)$ exist and $\lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow a^-} f(x)$.

This was because, in $\mathbb{R}$ one can approach a point either from right or left.

However, things are a bit more complicated in $\mathbb{R}^2$.



Suppose that $f(x, y) \rightarrow L_1$ as $(x, y) \rightarrow (a, b)$ along a path C_1 and $f(x, y) \rightarrow L_2$ as $(x, y) \rightarrow (a, b)$ along a path C_2 . If $L_1 \neq L_2$, then $\lim_{(x,y) \rightarrow (a,b)} f(x, y)$ does not exist.

This phenomena is sometimes called as **path independence of the limit**.

Definition We say that a function $f(x, y)$ is continuous at (a, b) if

$$\lim_{(x,y) \rightarrow (a,b)} f(x, y) = f(a, b).$$

Every polynomial, exponential function is continuous. Trigonometric functions, and logarithmic functions are continuous in their domains. Rational functions are continuous when their denominators are nonzero. Furthermore, a sum, difference, product or ratio* of continuous functions is continuous.

A polynomial in two variables is a linear combination of the terms of the form $cx^m y^n$ where $c \in \mathbb{R}$ and $n, m \in \mathbb{N}$.

Example 1. Evaluate the following limits, if they exist.

- $\lim_{(x,y) \rightarrow (3,-2)} (2x^2y - 5xy^3 + 41)$
- $\lim_{(x,y) \rightarrow (1,-1)} (e^{x-y} - 2 \cos(3x + y) + \sin(2y))$

Solution.

Since $f(x, y) = 2x^2y - 5xy^3 + 41$ is a continuous function (at $(2, 1)$), we have

$$\begin{aligned}\lim_{(x,y) \rightarrow (3,-2)} (2x^2y - 5xy^3 + 41) &= 2 \cdot 3^2 \cdot (-2) - 5 \cdot 3 \cdot (-2)^3 + 41 \\ &= 125.\end{aligned}$$

Example 2. For the following limits approach $(0,0)$ along the x -axis, and along the y -axis, and along a line passing through the origin. Does the limit exist?

► $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2}{x^2 + y^2} \quad (*)$

Along x -axis $y=0$

$$(*) = \lim_{x \rightarrow 0} \frac{x^2}{x^2} = 1$$

Along y -axis $x=0$

$$(*) = \lim_{y \rightarrow 0} \frac{0}{y^2} = 0$$

Along $y=mx$

$$(*) = \lim_{x \rightarrow 0} \frac{x^2}{x^2 + m^2 x^2} = \frac{1}{1+m^2}$$

$$(*) = \text{dne}$$

► $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 + \sin^2 y}{2x^2 + y^2} \quad (*)$

Along x-axis $y=0$

$$(*) = \lim_{x \rightarrow 0} \frac{x^2}{2x^2} = \frac{1}{2}$$

Along y-axis $x=0$

$$(*) = \lim_{y \rightarrow 0} \left(\frac{\sin y}{y} \right)^2 = 1^2 = 1$$

Along $y=mx$

$$(*) = \lim_{x \rightarrow 0} \frac{x^2 + \sin^2(mx)}{2x^2 + m^2 x^2} \stackrel{(\frac{0}{0})}{=} \lim_{x \rightarrow 0}$$

$$\frac{2x + \overbrace{2\sin(mx)\cos(mx)m}^{\sin(2mx)}}{4x + 2m^2 x} \quad \left(\frac{0}{0} \right)$$

$$\stackrel{L}{=} \lim_{x \rightarrow 0} \frac{2 + \cos(2mx) \cdot 2m^2}{4 + 2m^2} = \frac{2 + 2m^2}{4 + 2m^2}$$

$$(*) = \text{dne}$$

► $\lim_{(x,y) \rightarrow (0,0)} \frac{xy \cos y}{3x^2 + y^2}$

Clearly, the value of the limit is 0 along the x or the y-axis.

Along $y=mx$

$$(*) = \lim_{x \rightarrow 0} \frac{x \cdot mx \cdot \cos(mx)}{3x^2 + m^2x^2} = \lim_{x \rightarrow 0} \frac{m \cancel{x^2} \cos(mx)}{\cancel{x^2} (3 + m^2)}$$

$$= \frac{m \cos(0)}{3 + m^2} = \frac{m}{3 + m^2} \rightarrow \text{depends on } m$$

the limit DNE, so

► $\lim_{(x,y) \rightarrow (0,0)} \frac{6x^3y}{2x^4 + y^4}$

Clearly, the value of the limit along the x-axis and the y-axis is 0. However,

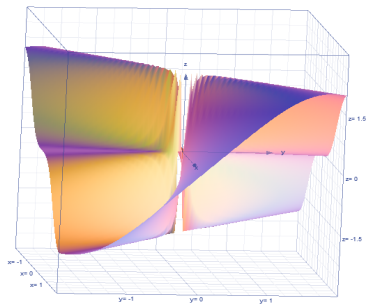
Along $y=mx$

$$(*) = \lim_{x \rightarrow 0} \frac{6x^3(mx)}{2x^4 + (mx)^4} = \lim_{x \rightarrow 0} \frac{6mx^4}{x^4(2+m^4)}$$

$$= \frac{6m}{2+m^4}, \text{ depends on } m.$$

Thus, $(*)$ d.n.e.

Note that the graph of $z = \frac{6x^3y}{2x^4+y^4}$ looks like:



Example 3. Find $\lim_{(x,y) \rightarrow (0,0)} \frac{xy^2}{x^2 + y^4}$. (*)

Solution.

Along x-axis $y \rightarrow 0$

$$(*) = \lim_{y \rightarrow 0} \frac{0}{x^2} = 0$$

Along $x=y^2$

$$(*) = \lim_{y \rightarrow 0} \frac{y^2 \cdot y^2}{(y^2)^2 + y^4} = \lim_{y \rightarrow 0} \frac{y^4}{2y^4} = 1/2$$

$0 \neq 1/2$

(*) = dne

Example 4. Evaluate $\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{\sqrt{x^2 + y^2}}$ (*)

Solution.

$$0 \leq \left| \frac{xy}{\sqrt{x^2 + y^2}} \right| = |x| \sqrt{\frac{y^2}{x^2 + y^2}} \leq |x|$$


Hence (*) $\Rightarrow$ by the squeeze theorem.

Example 4. cont'd Evaluate $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 \sin^2 y}{x^2 + 2y^2} \quad (*)$

Solution.

$$0 \leq \frac{x^2 \sin^2 y}{x^2 + 2y^2} \leq \sin^2 y$$

Diagram illustrating the squeeze theorem application. The expression $\frac{x^2 \sin^2 y}{x^2 + 2y^2}$ is circled, with a note ≤ 1 above it. Arrows point from the circled expression and $\sin^2 y$ towards the limit 0 . A note $\text{as } y \rightarrow 0$ is written near the right arrow.

Hence, $(*) = 0$ by the squeeze thm

Example 4. cont'd Evaluate $\lim_{(x,y) \rightarrow (0,0)} \frac{4x^2y}{x^2 + y^2}$ (*)

Solution.

$$(*) = 0$$

by the squeeze thm.

Remark. The computation above shows that

$$\lim_{(x,y) \rightarrow (0,0)} \left| \frac{4x^2y}{x^2 + y^2} \right| = 0.$$

But, this implies that

$$\lim_{(x,y) \rightarrow (0,0)} \frac{4x^2y}{x^2 + y^2} = 0.$$

as well.

Example 4. cont'd Evaluate $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y \cos(\frac{2}{x^3})}{x^2 + 5y^2}$ (*)

$$\left| \frac{x^2 y \cos(\frac{2}{x^3})}{x^2 + 5y^2} \right| \leq \left| y \overbrace{\cos(\frac{2}{x^3})}^{\leq 1} \right| \leq |y| \xrightarrow[y \rightarrow 0]{\text{as}} 0$$

$$\Rightarrow \lim_{(x,y) \rightarrow (0,0)} \left| \frac{x^2 y \cos(2/x^3)}{x^2 + 5y^2} \right| = 0, \text{ we must have}$$

(*) = 0 as well.

Example 5. Evaluate $\lim_{(x,y,z) \rightarrow (0,0,0)} \frac{x^2 + 2y^2 + 3z^2}{x^2 + y^2 + z^2}$. (*)

Solution.

(*) dne as

Along x-axis $y=z=0$

$$(*) = \lim_{x \rightarrow 0} \frac{x^2}{x^2} = 1$$

Along y-axis $x=z=0$

$1 \neq 2$.

$$(*) = \lim_{y \rightarrow 0} \frac{2y^2}{y^2} = 2$$

Example 6. Determine the value(s) of c so that the following function is continuous at $(0,0)$.

$$f(x,y) = \begin{cases} \frac{x^2 y^3}{2x^2 + y^2}, & \text{if } (x,y) \neq (0,0) \\ c-1, & \text{if } (x,y) = (0,0) \end{cases}$$

For continuity, we must have $\lim_{(x,y) \rightarrow (0,0)} f(x,y) = f(0,0)$.

$$f(0,0) = c-1.$$

Note that,

$$\left| \frac{x^2 y^3}{2x^2 + y^2} \right| \leq |y^3| \rightarrow 0 \quad \text{as } y \rightarrow 0$$

$$\text{Thus, } \lim_{(x,y) \rightarrow (0,0)} \left| \frac{x^2 y^3}{2x^2 + y^2} \right| = 0 \Rightarrow \lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y^3}{2x^2 + y^2} = 0$$

$$\Rightarrow c-1=0 \Rightarrow c=1.$$

Example 7. Determine whether the given function is continuous at $(0,0)$ or not.

$$f(x,y) = \begin{cases} \frac{xy}{x^2+xy+y^2}, & \text{if } (x,y) \neq (0,0) \\ 0, & \text{if } (x,y) = (0,0) \end{cases}$$

Let us approach $(0,0)$ by straight lines passing through $(0,0)$
 $y = mx \Rightarrow$

$$\begin{aligned} \lim_{(x,y) \rightarrow (0,0)} f(x,y) &= \lim_{x \rightarrow 0} \frac{x \cdot mx}{x^2 + x \cdot mx + m^2 x^2} = \lim_{x \rightarrow 0} \frac{m x^2}{x^2(1+m+m^2)} \\ &= \frac{m}{1+m+m^2} \rightarrow \text{dne so } \lim_{(x,y) \rightarrow (0,0)} f(x,y) \text{ DNE} \end{aligned}$$

Thus, $f(x,y)$ is discont. at $(0,0)$

Example 8. Determine the set of points at which the function is continuous.

$$f(x, y) = \frac{\cos(x - y)}{e^x - y^2}, \quad g(x, y) = \ln(2 + 3x^2 + 4y^6) + \sqrt{x + y^2}$$

f is $\swarrow$ cont. when $e^x - y^2 \neq 0 \Rightarrow y^2 \neq e^x \Rightarrow y \neq \pm e^{x/2}$

g is cont. when

$$2 + 3x^2 + 4y^6 > 0 \text{ and } x + y^2 \geq 0 \Rightarrow x \geq -y^2$$

$\hookrightarrow$ true for all $(x, y) \in \mathbb{R}^2$

Example 9. Show that $\lim_{(x,y) \rightarrow (0,0)} \frac{x^5 + y^5}{x^4 + y^4} = 0$.

Hint. Write

$$\frac{x^5 + y^5}{x^4 + y^4} = \frac{x^4 x + y^4 y}{x^4 + y^4} = \frac{x^4 x}{x^4 + y^4} + \frac{y^4 y}{x^4 + y^4}$$

Then, split the given limit into two limits, and use the squeeze theorem for each one.

We may want to try **polar coordinates**

$$x = r \cos(\theta), \quad y = r \sin(\theta)$$

when we investigate limit problems. Notice that as $(x, y) \rightarrow (0, 0)$, we must have $r \rightarrow 0$.

Example 10. Find $\lim_{(x,y) \rightarrow (0,0)} \frac{e^{-x^2-y^2} - 1}{x^2 + y^2} \quad (*)$

Answer.

$$x = r \cos \theta, \quad y = r \sin \theta \Rightarrow$$

$$(*) = \lim_{r \rightarrow 0} \frac{e^{-r^2} - 1}{r^2} \stackrel{L}{=} \lim_{r \rightarrow 0} \frac{e^{-r^2} \cdot (-2r)}{2r}$$

$$= - \lim_{r \rightarrow 0} e^{-r^2} = -e^0 = -1$$

Example II. Find $\lim_{(x,y) \rightarrow (0,0)} (x^2 + y^2) \ln(x^2 + y^2)$ (*)

Answer.

Using the polar coordinates $x = r \cos \theta$, $y = r \sin \theta$
we have $\rightarrow (0, \infty)$ type of limit

$$(*) = \lim_{r \rightarrow 0} r^2 \ln(r^2) = \lim_{r \rightarrow 0} \frac{\ln(r^2)}{\frac{1}{r^2}}$$

$$\stackrel{L}{=} \lim_{r \rightarrow 0} \frac{\frac{2r}{r^2}}{\frac{-2}{r^3}} = \lim_{r \rightarrow 0} \frac{2r}{r^2} \left(\frac{-r^3}{2} \right)$$

$$= \lim_{r \rightarrow 0} (-r^2) = 0.$$

Remark. Consider

$$\lim_{(x,y) \rightarrow (0,0)} \frac{xy^2}{x^2 + y^4} \quad (*)$$

By switching to the polar coordinates we obtain

$$\lim_{r \rightarrow 0} \frac{r \cos(\theta) \sin^2(\theta)}{\cos^2(\theta) + r^2 \sin^4(\theta)} \quad (**)$$

Clearly, the value of this limit is 0 for each **fixed** value of θ .

However, $(*)$ dne as along the parabola $x = y^2$,

$(*)$ becomes

$$\lim_{y \rightarrow 0} \frac{y^2 y^2}{y^4 + y^4} = \frac{1}{2}$$

But along the x -axis the limit value is 0.

Additional Examples

Example 12. Evaluate $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 \tan(y) - y^2 \tan(x)}{x^2 + y^2}$

Solution.

$$\begin{aligned} \lim_{(x,y) \rightarrow (0,0)} \frac{x^2 \tan y - y^2 \tan x}{x^2 + y^2} &= \lim_{(x,y) \rightarrow (0,0)} \frac{x^2 (\tan y - y^2) - y^2 (\tan x - x^2)}{x^2 + y^2} \\ &= \lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{x^2 + y^2} \cdot \frac{(\tan y - y^2)}{y} - \lim_{(x,y) \rightarrow (0,0)} \frac{y^2 x}{x^2 + y^2} \cdot \frac{(\tan x - x^2)}{x}. (*) \end{aligned}$$

On the other hand,

$$\lim_{y \rightarrow 0} \frac{\tan y - y^2}{y} \stackrel{(L)}{=} \lim_{y \rightarrow 0} \frac{\sec^2 y - 2y}{1} = 1, \text{ and}$$

$$0 \leq \left| \frac{x^2 y}{x^2 + y^2} \right| \leq |y| \Rightarrow \lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{x^2 + y^2} = 0$$

↙ as $y \rightarrow 0$ ↘

Thus,

$$(*) = 0 \cdot 1 - 0 \cdot 1 = 0$$

Example 13. Evaluate $\lim_{(x,y) \rightarrow (0,0)} \frac{x \sin(2y) - y \sin(2x)}{x^4 + y^4}$ (*)

Solution.

Let us approach (0,0) along the path $y = mx \Rightarrow$

$$(*) = \lim_{x \rightarrow 0} \frac{x \sin(2mx) - mx \sin(2x)}{x^4 + m^4 x^4}$$

$$= \lim_{x \rightarrow 0} \frac{\sin(2mx) - m \sin(2x)}{(1+m^4) x^3}$$

$$= - \frac{4m(m^2-1)}{3(1+m^4)} \quad \text{using the L'Hopital's Rule}$$

, depends on m

Thus, (*) dne.

Example 14. Find the value of α so that

$$f(x, y) = \begin{cases} \frac{(x^2 - y^2)^2}{x^2 + y^2}, & \text{if } (x, y) \neq (0, 0) \\ \alpha - 9, & \text{if } (x, y) = (0, 0) \end{cases}$$

is continuous at $(0, 0)$.

Solution.

$$0 \leq \left| \frac{(x^2 - y^2)^2}{x^2 + y^2} \right| = \frac{(x^2 - y^2)^2}{x^2 + y^2} \leq \frac{(x^2 + y^2)^2}{x^2 + y^2} = x^2 + y^2$$

$\searrow \quad \swarrow$
 $(x, y) \rightarrow (0, 0)$

$$\text{Thus, } \lim_{(x, y) \rightarrow (0, 0)} \left| \frac{(x^2 - y^2)^2}{x^2 + y^2} \right| = 0 \quad \Rightarrow$$

$$\lim_{(x, y) \rightarrow (0, 0)} \frac{(x^2 - y^2)^2}{x^2 + y^2} = 0.$$

Thus, for the continuity of $f(x, y)$, we must have

$$\lim_{(x, y) \rightarrow (0, 0)} f(x, y) = f(0, 0)$$

$$0 = \alpha - 9 \quad \Rightarrow \quad \boxed{\alpha = 9}$$

Example 15. Evaluate $\lim_{(x,y) \rightarrow (-3,2)} \frac{2x^2 + 5xy + 3y^2}{2x + 3y}$

Solution.

$$\begin{aligned} \lim_{(x,y) \rightarrow (-3,2)} \frac{2x^2 + 5xy + 3y^2}{2x + 3y} &= \lim_{(x,y) \rightarrow (-3,2)} \frac{\cancel{(2x+3y)} (x+y)}{\cancel{(2x+3y)}} \\ &= \lim_{(x,y) \rightarrow (-3,2)} (x+y) = -3 + 2 = -1. \end{aligned}$$

Example 16. Evaluate $\lim_{(x,y) \rightarrow (1,0)} (1+xy)^{1/xy}$

Solution.

$$\text{Let } t := xy \Rightarrow (x,y) \rightarrow (1,0) \Leftrightarrow t \rightarrow 0$$

$$\text{So the given limit becomes } \lim_{t \rightarrow 0} (1+t)^{1/t}$$

$$\text{Let } y := (1+t)^{1/t} \Rightarrow \ln y = \frac{\ln(1+t)}{t} \Rightarrow$$

$$\lim_{t \rightarrow 0} \frac{\ln(1+t)}{t} \stackrel{L}{=} \lim_{t \rightarrow 0} \frac{\frac{1}{1+t}}{1} = 1 \Rightarrow$$

$$\lim_{t \rightarrow 0} (1+t)^{1/t} = e^1 = e$$

Example 17. Evaluate $\lim_{(x,y) \rightarrow (0,0)} \frac{(x^2 + y^2)^2}{x^4 + y^4}$

Solution.

Along $y=mx$, the given limit becomes

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{(x^2 + m^2 x^2)^2}{x^4 + m^4 x^4} &= \lim_{x \rightarrow 0} \frac{x^4 (1+m^2)^2}{x^4 (1+m^4)} \\ &= \frac{(1+m^2)^2}{1+m^4}, \quad \text{depends on } m \end{aligned}$$

so the limit does not exist.

Example 18. Find $a, b \in \mathbb{R}$, such that

$$f(x, y) = \begin{cases} \frac{x^2(y-1)}{3x^2 + 7(y-1)^2} + a, & \text{if } x > 0, \\ \frac{\tan^2(x+y-1)}{(x+y-1)^2} + b, & \text{if } x < 0, \\ 9, & \text{if } x = 0, \end{cases}$$

is continuous at the point $(0, 1)$.

Solution. For simplicity, let us make a change of a variable: $Y := y - 1$. Then we have

$$f(x, Y) = \begin{cases} \frac{x^2 Y}{3x^2 + 7Y^2} + a, & \text{if } x > 0, \\ \frac{\tan^2(x+Y)}{(x+Y)^2} + b, & \text{if } x < 0, \\ 9, & \text{if } x = 0, \end{cases}$$

Clearly, we have

$$\lim_{(x,y) \rightarrow (0,1)} f(x, y) = \lim_{(x,Y) \rightarrow (0,0)} f(x, Y).$$

For $x < 0$, we have

$$\begin{aligned} \lim_{(x,Y) \rightarrow (0,0)} f(x, Y) &= \lim_{(x,Y) \rightarrow (0,0)} \frac{\tan^2(x+Y)}{(x+Y)^2} + b \\ &= \lim_{(x,Y) \rightarrow (0,0)} \frac{\tan^2(\theta)}{\theta^2} + b \quad (\theta = x+Y) \\ &= 1 + b \end{aligned}$$

On the other hand,
For $x > 0$, we have

$$\lim_{(x,Y) \rightarrow (0,0)} f(x,Y) = \lim_{(x,Y) \rightarrow (0,0)} \frac{x^2 Y}{3x^2 + 7Y^2} + a$$

Since

$$\left| \frac{x^2 Y}{3x^2 + 7Y^2} \right| \leq |Y| \rightarrow 0 \quad (\text{as } Y \rightarrow 0)$$

we conclude that

$$\lim_{(x,Y) \rightarrow (0,0)} f(x,Y) = \lim_{(x,Y) \rightarrow (0,0)} \frac{x^2 Y}{3x^2 + 7Y^2} + a = 0 + a = a.$$

In order for f to be continuous at $(x,y) = (0,1)$, we must have

$$\lim_{(x,y) \rightarrow (0,1)} f(x,y) = f(0,1)$$

$$\implies$$

$$1 + b = a = 9 \implies$$

$$a = 9 \quad \text{and} \quad b = 8.$$